

Dr. Babasaheb Ambedkar Technological University, Lonere

Supplementary Winter Semester Examination – 2025

Course: B.Tech. (Computer Engineering / Computer Science and Engineering / Computer Science and Design / Artificial Intelligence and Allied)

Subject Code & Name: BTES405 Digital Logic Design and Microprocessor

Semester : IV

Max Marks: 60

Date: 25th January 2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- | | | |
|----|--|---|
| 1 | The 8086 microprocessor has a data bus width of _____.
a) 16 bits b) 8 bits c) 32 bits d) 64 bits | 1 |
| 2 | A bistable device capable of storing one bit of information is called a _____.
a) Flip-flop b) Register c) Counter d) Decoder | 1 |
| 3 | Which instruction category affects the flow of program execution?
a) Logical b) Arithmetic c) Data transfer d) Branch | 1 |
| 4 | The logic gate whose output is LOW only when all inputs are HIGH is the _____ gate.
a) NAND b) NOR c) XOR d) OR | 1 |
| 5 | A ring counter is a type of _____ shift register.
a) Bidirectional b) Universal c) Circular d) Parallel | 1 |
| 6 | A Boolean function expressed as a sum of minterms is called _____ form.
a) Canonical SOP b) Canonical POS c) Standard POS d) Reduced SOP | 1 |
| 7 | The instruction used to call a subroutine in 8086 is _____.
a) JMP b) RET c) CALL d) INT | 1 |
| 8 | The function of the program counter is to _____.
a) Store operands b) Hold the address of the next instruction
c) Manage I/O devices d) Control interrupts | 1 |
| 9 | The number system most suitable for representing signed numbers in computers is _____.
a) Signed magnitude b) One's complement
c) Two's complement d) Excess-3 | 1 |
| 10 | The circuit that selects one input from many and forwards it to a single output is _____.
a) Decoder b) Encoder c) Multiplexer d) Demultiplexer | 1 |

11	DMA allows data transfer between memory and I/O devices _____.		1
	a) Using CPU instructions b) Without CPU intervention		
	c) Through interrupts only d) Via ALU operations		
12	The hexadecimal equivalent of binary number 1110 is _____.		1
	a) E b) D c) C d) F		
Q. 2	Solve the following.		12
A)	Define digital signals and list two advantages over analog signals. State the truth tables of AND, OR, and NOT gates.	CO1	6
B)	Write the basic postulates of Boolean algebra. List different number systems used in digital electronics.	CO1	6
Q.3	Solve the following.		12
A)	State the purpose of Karnaugh maps. Explain the concept of don't-care conditions in K-maps.	CO2	6
B)	Define multiplexer and demultiplexer. Describe the working of a 4:1 multiplexer.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Define latch and flip-flop. Explain the working of SR flip-flop with clock.	CO3	6
B)	List different types of flip-flops. Describe the operation of JK flip-flop.	CO3	6
C)	Difference between synchronous and asynchronous counters.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Define microprocessor. Explain the internal architecture of 8086.	CO4	6
B)	Describe the role of registers in CPU operation.	CO4	6
C)	List the features of 8086 microprocessor. Explain in detail the memory organization of 8086.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	List the addressing modes of 8086. Explain any FOUR addressing modes with examples.	CO5	6
B)	Describe the interrupt handling mechanism in 8086.	CO5	6
C)	State the purpose of assemblers. Explain the difference between assembler and compiler.	CO5	6

*** End ***